

- ① Coleman Country has a phone exchange that handles calls between callers and those receiving the call. Given these three actors, draw a simple sequence diagram for making a simple phone call.

Ans.: For a simple phone call, we have three actors:

- ① Caller
- ② Phone Exchange (Coleman Country)
- ③ Receiver

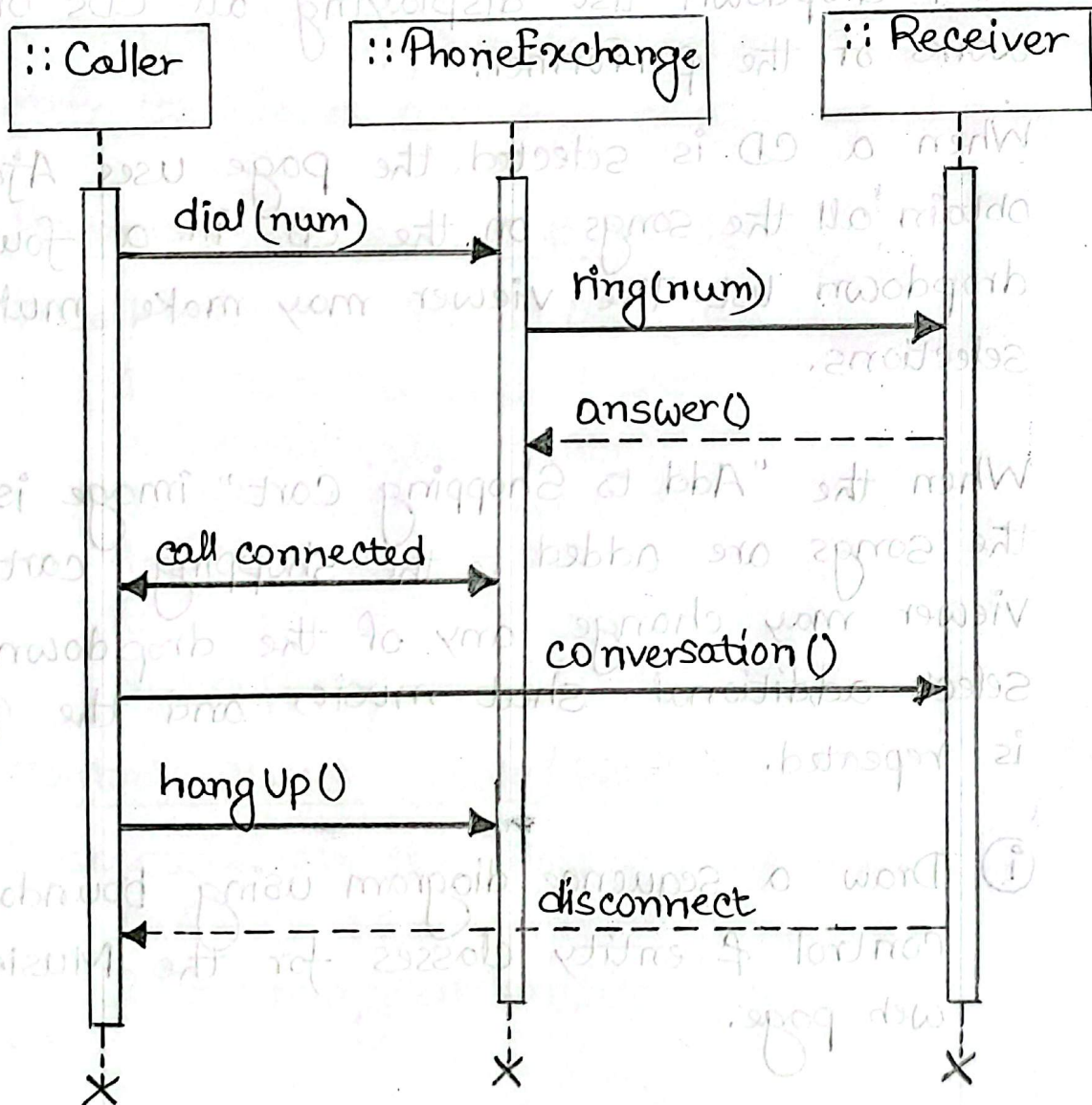


figure 1.1: Sequence diagram for making a simple phone call.

- ② Musixscore.com is an online service providing sheet music to customers. On their "Browse Music" page, customers select a genre of music from a dropdown list. The webpage uses Ajax to obtain a list of performers, musicians, or groups that match the genre, which is formatted as a dropdown list.

When a selection is made from the performer's dropdown list, the page uses Ajax to display a third dropdown list displaying all CDs or other works of the performer.

When a CD is selected, the page uses Ajax to obtain all the songs on the CD in a fourth dropdown list. The viewer may make multiple selections.

When the "Add to Shopping Cart" image is clicked, the songs are added to the shopping cart. The viewer may change any of the dropdown lists to select additional sheet music, and the process is repeated.

- ① Draw a sequence diagram using boundary, control & entity classes for the Musixscore web page.

- ② Write a list of messages, names, and parameters, along with data types that would be passed to the classes and the values (with data types) that are included with the return message.
- ③ Create a class diagram for the entity classes used in the sequence diagram.

Ans.: ① From the given scenario, we are creating an ECB Sequence Diagram. In order to do that, we identify the classes according to the E-C-B model:

□ Boundary Classes (UI & Interaction)

These handle user interaction (web interface & ajax)

1. Browse Music Page (:BMP)
2. GenreDropDown (:GDD)
3. PerformerDropDown (:PDP)
4. CD Drop Down (:CDD)
5. Song DropDown (:SD)
6. ShoppingCart UI (:SCI)

□ Control Classes (Logic/Work-flow)

These coordinate the flow & Ajax requests:

1. Browse Music Controller (:BMC)
2. ShoppingCart Controller (:SCC)

Entity Classes (Data/Business Objects)

These represent persistent domain data :

1. Genre (:GN)
2. Performer (:PR)
3. CD (:CD)
4. Song (:SN)
5. ShoppingCart (:SC)

Boundary Classes (UI & Interaction)

1. BrowseMusicPage (:BMP)
2. GenreDropDown (:GD)
3. PerformerDropDown (:PD)
4. CDListDisplay (:CDL)
5. SongListDisplay (:SLD)

Control Classes (Logic/Work-flow)

1. BrowseMusicController (:BMC)
2. ShoppingCartController (:SCC)

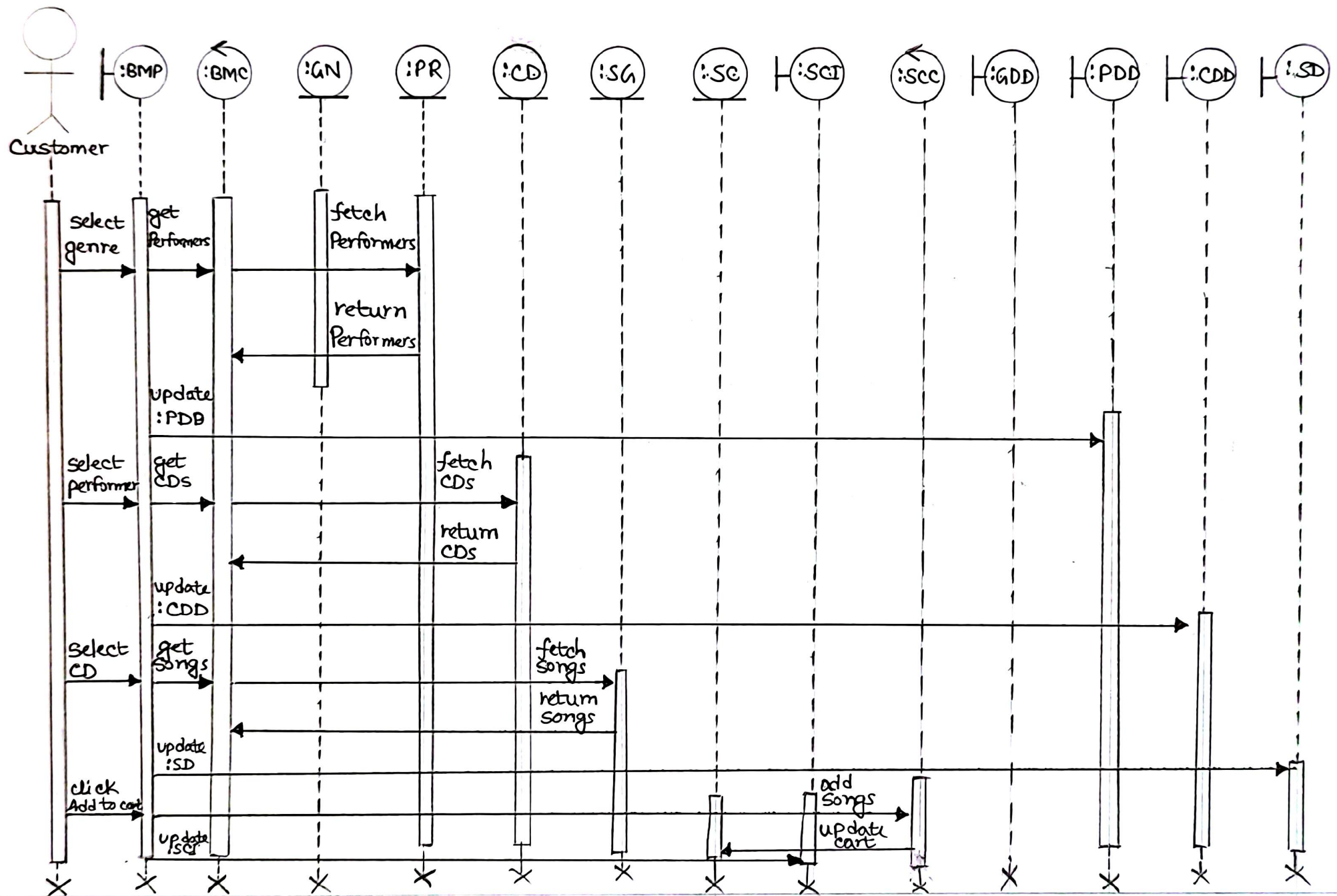


Figure 2.1: ECB Sequence Diagram of the scenario.